Collisions in two dimensions

AS Level | 9702 AS Ch.6 | Chapter 6: Momentum | Lesson 6.5 | Sections 6.5

Goal: Solve a glancing collision by conserving momentum in two perpendicular directions.

velocity

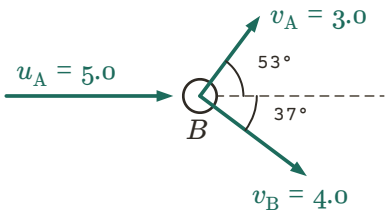
01 The big picture

Momentum is a vector, so in a glancing collision it is conserved in every direction at once. Choose two perpendicular directions (x along the original motion, y across it) and conserve momentum in each separately.

Resolve each momentum into components with $\cos$ and $\sin$. If nothing was moving in the y direction before, the y-components after must cancel.

Glancing collision [paraphrase]
A collision that is not head-on: the objects move off at angles.

Fig. 1 · A glancing collision



A and B move off at angles; their y-momenta cancel and their x-momenta add up to A's momentum before.

02 Key equations

Along x

$$\sum p_x \text{ before} = \sum p_x \text{ after}$$

Along y

$$\sum p_y \text{ before} = \sum p_y \text{ after}$$

Components

$$p_x = p\cos\theta, \quad p_y = p\sin\theta$$

Symbols and units

p_x	momentum along x	$\mathrm{kg\,m\,s^{-1}}$	θ	angle to the x direction	$^\circ$

03 Sample problems with solutions

Sample 1 · A glancing collision
A 2.0 kg puck A at 5.0 m s^{-1} hits an identical puck B at rest. A leaves at 3.0 m s^{-1} at 53° to its first direction (Fig. 1). Find B's velocity.

Step 1. Along x (masses equal, so they cancel):

$$5.0 = 3.0\cos 53^\circ + v_x \;\Rightarrow\; v_x = 3.2\ \mathrm{m\,s^{-1}}$$

Step 2. Along y (zero before):

$$0 = 3.0\sin 53^\circ + v_y \;\Rightarrow\; v_y = -2.4\ \mathrm{m\,s^{-1}}$$

Step 3. Combine:

$$v = \sqrt{3.2^2 + 2.4^2} = 4.0\ \mathrm{m\,s^{-1}}, \quad \theta = \tan^{-1}\frac{2.4}{3.2} = 37^\circ$$

Answer: 4.0 m s^{-1} at 37° on the other side of the original direction

04 Practice problems

P1 In a two-dimensional collision, what is conserved separately?
Answer: Momentum in each of two perpendicular directions

P2 A 1.0 kg ball moving east at 4.0 m s^{-1} sticks to a 1.0 kg ball moving north at 4.0 m s^{-1}. Find their velocity.
Answer: 2.8 m s^{-1} north-east (45°)

P3 A 3.0 kg object at rest splits into three 1.0 kg pieces; two go east and north at 6.0 m s^{-1}. Find the third.
Answer: 8.5 m s^{-1} south-west

P4 Is the collision in the sample elastic?
Answer: Yes: 25 J of kinetic energy before and after

05 Other useful information

Common mistakes

✗ **Wrong:** Add the speeds after: 3.0 + 4.0 = 5.0?
✓ **Right:** Add momentum components, never speeds.

Exam tips

- Draw the before and after momenta tip to tail: they close a triangle.

What the syllabus asks

- 9702 topic 3: conservation of momentum in one and two dimensions. Coursebook 6.5.

Explore: Practice questions for this lesson